Spin-based $\begin{cases} \text{classical} \\ \text{AQC} \end{cases}$

graph G, $\underline{\sigma} = (\sigma_1, \sigma_2, \ldots, \sigma_N)$

$\sigma = \begin{cases} +1 & \text{one part} \\ -1 & \text{another part.} \end{cases}$

$C(\underline{\sigma})$ cut-function. $\}$ $C(\underline{\sigma}) = \frac{W}{2} - \frac{H(\underline{\sigma})}{2}$

$H(\underline{\sigma})$ Hamiltonian $\}$

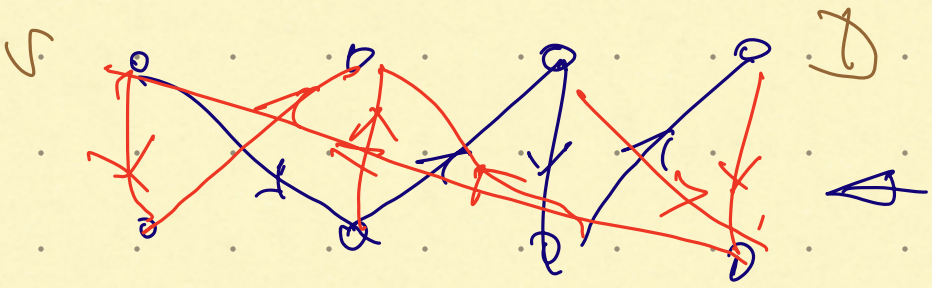
Maximum cut = ground state

LS hierarchy $\left(\underline{\underline{k\text{-opt.}}}\right)_k$ $O(e^k)$

Simulated annealing

Approximation ratio $\rho = \frac{C_{worst}}{C_g} \geq \frac{1}{2}$

as $N \to \infty$ $\rho_{LS} \to \rho_{random}$



$\not\Delta$ edges $\rightarrow$ spins.

$\downarrow \Rightarrow \sigma = -1$ $\qquad$ $\uparrow \Rightarrow \sigma = +1$.

$$G(\underline{\sigma}) = \sum_{i \in V_2} Q_i^2(\sigma)$$

$Q_i(\sigma) \in \{0, \pm 2, S, D\}$

$$Q_i(\sigma) = \sum_m \sigma_m + p \cdot \sigma_0$$

$\uparrow$ edges incident to i.

σ^* : $\boxed{G(\sigma^*) = 0}$.

Path = node σ_n^* that flipped

That $\sigma_n^* = -\sigma_n^{(0)}$

Hypoth: All LS-algos will fail to do this."

$\Leftarrow$ $\#(\text{zeroes}) \sim e^{N-k}$ "

N - number of nodes.

BFS / DFS $\rightarrow O(N^2)$

Problem about path.

Answer: V_2 - model.

Because, it's not limited by Hamming distance.

σ | σ' $\quad d_H(\sigma, \sigma')$ is arbitrary

1 1 1 } $d_H = 1$
1 0 0 }

1 1 1 } $d_H = 2$
1 0 0 }

(a) k-soft is $\begin{cases} d_H(\sigma, \sigma') \leq k \\ d_H(\ ) \text{ is unbounded.} \end{cases}$

$\nexists\ V_2$

(b) Not rounding-unitive

(c) Multiple solutions.

V_2 model

$\psi = (\sigma, x)$ - relaxed spin

$\sigma \in \{-1, 1\} \qquad x \in [-1, 1]$

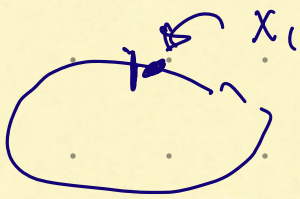
$\psi \in \{-1, 1\} \times [-1, 1] / \sim$

(F) $(\sigma, 1) \sim (-\sigma, -1)$



$$C_{V_2}(T) = C(0) + \frac{1}{4} \sum_{m,n} A_{m,n} \sigma_m \sigma_n |x_m - x_n|$$

$$C(0) = \frac{1}{4} \sum_{m,n} A_{mn} (1 - \sigma_m \sigma_n).$$



when $x_1 = l, \sigma_1 = \sigma_1$

$x_1 = -l \quad \sigma_1 = -\sigma_1$

$$C(0) + \frac{1}{4} \sum_{m=1,n} A_{mn} \sigma_m \sigma_n |x_1 - x_n|$$

$$+ \frac{1}{4} \sum_{m,n=1} A_{m,n} \sigma_m \sigma_n |x_m - x_1|$$

$$= C(0) + \sum_{m \in (1, n \neq 1} \cdots$$

$$= C(\sigma) + \frac{1}{2} \sum_{n} A_{1n} \sigma_1 \sigma_n |x_1 - x_n|$$

$C(\psi)$ $\quad = \quad$ $\underline{1}$.

$$+ \sum_{n \neq 1, n \neq 1} \cdots$$

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$$C(\psi') = C(\sigma | \sigma_1 \to -\sigma_1) - \frac{1}{2} \sigma_1 \sum A_{1n} \sigma_n |-1 - x_n|$$

$$+ \sum_{n \neq 1, n \neq 1} \cdots$$

$$C(\sigma) = \frac{1}{4} \sum A_{m,n} (1 - \sigma_m \sigma_n)$$

$$= \frac{1}{4} \sum_{m=1, n} A_{m,n} (1 - \sigma_1 \sigma_n)$$

$$+ \frac{1}{4} \sum_{m, n=1} A_{mn} (1 - \sigma_m \sigma_1)$$

$$+ \frac{1}{4} \sum_{m \neq 1, n \neq 1} \cdots$$

$$= \frac{1}{2} \sum_{m=1} A_{1n} (1 - \sigma_1 \sigma_n) + \cdots \quad n \neq 1.$$

$$C(\psi) = \frac{1}{2}\sum A_{1,n}(1-\sigma_1\sigma_n) + \dots$$

$$\| \quad ? \quad + \frac{1}{2}\sum A_{1,n}\sigma_1\sigma_n\,|1-\chi_n| + \dots$$

$$C(\psi') = \frac{1}{2}\sum A_{1,n}(1+\sigma_1\sigma_2) + \dots$$

$$-\frac{1}{2}\sum A_{1,n}\sigma_1\sigma_n\,|-1-\chi_n| + \dots$$

We need to prove

$$-\frac{1}{2}\sum A_{1,n}\sigma_1\sigma_n + \frac{1}{2}\sum A_{1,n}\sigma_1\sigma_n|1-\chi_n|$$

$$\overset{?}{=} \frac{1}{2}\sum A_{1,n}\sigma_1\sigma_n - \frac{1}{2}\sum A_{1,n}\sigma_1\sigma_n|-1-\chi_n|$$

$$|1-\chi_n| = 1-\chi_n$$

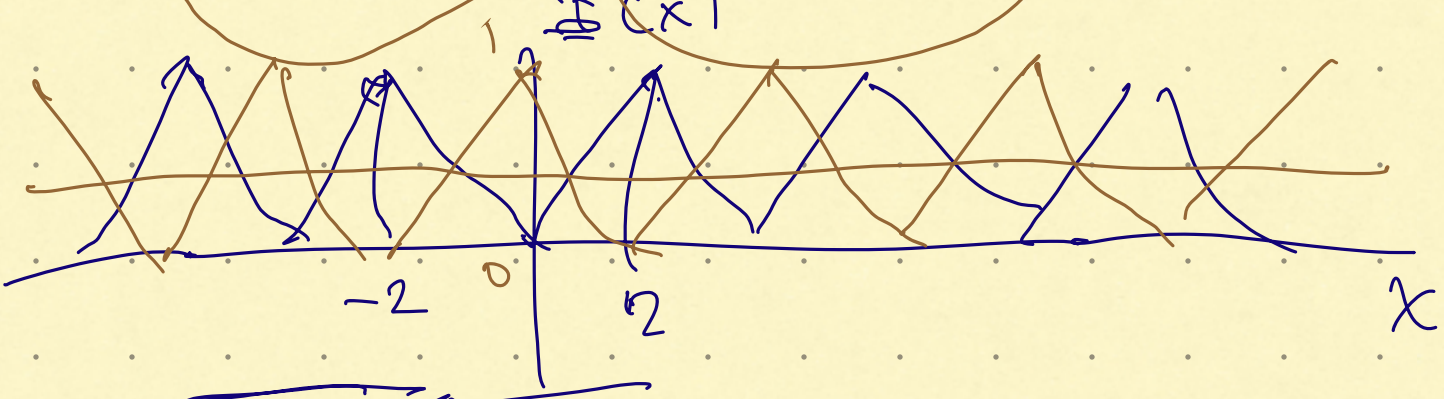
$$|-1-\chi_n| = |(-1)(1+\chi_n)| = 1+\chi_n$$

$$-\frac{1}{2}\sum A_{1,n}\sigma_1\sigma_n + \frac{1}{2}\sum A_{1,n}\sigma_1\sigma_n(1-\chi_n)$$

$$\overset{?}{=} \frac{1}{2}\sum A_{1,n}\sigma_1\sigma_n - \frac{1}{2}\sum A_{1,n}\sigma_1\sigma_n(1+\chi_n)$$

Indeed $\boxed{C(\psi) = C(\psi')}$.

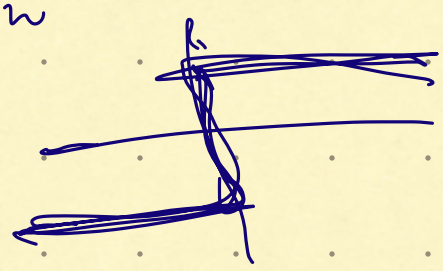
$$\Phi(x \pm \eta) = 1 - \Phi(x)$$



$$C_n(\psi) = C(\psi) + \frac{1}{2} \sum_m A_{nm} \sigma_m \sigma_n \Phi(x_m - x_n)$$

R-DIM

$$\dot{x}_n = \sum_m A_{nm} \sigma_m \sigma_n \operatorname{sign}(x_m - x_n).$$



$$C_{min} \le C_n(\psi) \le C_{max}$$